

A 67.355% Certificate for Simple Critical Zeros of the Riemann Zeta Function

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Abstract

We strengthen both components of the retained nine-point certificate: its global eight-point floor and its complete compatible-pair cover. Forty of 241 original captured frames receive replacement rational duals; one requires four closed branches. Ten compatible-pair domains receive twenty closed total-span branches. All 2,399 compatible domains are rechecked with exact rational arithmetic, including the 47 previously subdivided domains and every retained original dual. The resulting uniform reward is $805448/10^8$. The unchanged all-zero lossless transport gives the lower proportion

$$p = \frac{66812491}{99194552} = 0.6735500050446319 \dots > 0.67355$$

for simple critical-line zeros relative to all nontrivial zeros counted with multiplicity. The new finite certificate uses only original certified kernel points and convexity regions. Its mathematical proof and exact checker are presented here; subsequent Lean and independent kernel verification are recorded separately. No website acceptance or exhaustive global priority claim is made.

1 Statement and fixed framework

Let $N(T)$ count all nontrivial zeros of ζ in $(0, T]$ with multiplicity, and let $N_0^s(T)$ count simple zeros on $\Re s = 1/2$. The same superscripts on $N(T, 2T)$ denote the interval $(T, 2T]$.

Theorem 1.1. *Under the retained proved continuous AM certificate and its lossless all-zero analytic transport,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq p, \quad \liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq p, \quad p = \frac{66812491}{99194552}.$$

In particular the proportion 67.355% holds for all sufficiently large heights. Simple-zero inclusion implies the same lower bounds for distinct critical-line zeros.

Here “under” identifies the retained proved components of the certificate, not a new moment or correlation conjecture. We compare with the project’s preceding bound $668124910/991945881$ [3]:

$$100 \left(p - \frac{668124910}{991945881} \right) = \frac{602982731275}{24598906818510078} = 0.0000245125824380 \dots$$

percentage points. The strict crossing is exact: $p - 13471/20000 = 1251/247986380000 > 0$.

The unchanged kernel is $K(t) = \int_{-1/2}^{1/2} f(u) e^{-2\pi i t u} du$, with $w(t) = K(t)^2$, where the positive normalized density is

$$f(u) = \frac{\cos(\sqrt{2}u) + \sum_{j=1}^{12} c_j \cos(2\pi j u)}{\sqrt{2} \sin(1/\sqrt{2})}.$$

3 Exact finite certificates

We reconstruct the original YA lower bound for every captured cell. There are exactly forty original cells below the new floor. Their zero-based labels are

$$(0, 17, 19, 26, 33, 37, 46, 53, 61, 64, 67, 68, 69, 70, 71, 72, 73, 74, 78, \\ 110, 111, 120, 130, 131, 143, 144, 148, 151, 158, 159, 160, 163, 165, 166, \\ 173, 183, 184, 213, 215, 240).$$

Thirty-nine cells have one replacement dual each. Cell 17 uses four closed contiguous intervals of its first gap, with exact union equal to the original first-gap interval. All 201 remaining cells keep their original YA payment. Each problem has 33 physical variables: seven gaps and 26 kernel squares. The boxes of the squares remain the full $[0, 1]$ intervals. Across these 43 paid problems there are 449 added rows and 1,194 positive dual multipliers.

The ten new pair domains, in ordered cell labels, are

$$(389, 68), (309, 148), (69, 158), (399, 310), (401, 310), \\ (69, 160), (401, 309), (68, 160), (310, 401), (160, 69).$$

Each has two contiguous closed total-span branches. The original whole-domain inequalities and the branch inequalities determine the actual closure; there is no independent choice of left and right kernel values for their shared spans. These problems use 42 variables: eight gaps and 34 physical squares. Their 810 positive dual multipliers include 730 positive tangent rows.

All added tangents use 425 entries of the original 2,196-point PTL table and its retained REG convexity regions. Original point indices, value/derivative packs and source identity are frozen and checked. No newly sampled analytic data or curvature conjecture is used. If $v \leq w(p)$, $d_- \leq w'(p) \leq d_+$, and a certified convexity region contains $[l_*, u_*]$, with $l_* := \min(l, p)$ and $u_* := \max(u, p)$, then for $x \in [l, u]$,

$$w(x) \geq v + d_+(l_* - p) + d_-(x - l_*), \\ w(x) \geq v + d_-(u_* - p) + d_+(x - u_*).$$

These are genuine convex tangent bounds with derivative-interval corrections. The query interval and the anchor both lie in the certified region. An anchor outside the query interval is never used without the indicated corrections.

For the finite implication, write $Ax \leq b$, $L \leq x \leq U$ and the objective numerator as $q^T x$. For an exact nonnegative multiplier vector λ , set $r = q + A^T \lambda$. Then

$$q^T x \geq -\lambda^T b + \sum_i r_i \begin{cases} L_i, & r_i \geq 0, \\ U_i, & r_i < 0. \end{cases}$$

The bound includes every residual, even if numerically small. Scaling back gives the rational lower bound for the real functional. Floating optimization proposes witnesses only; it does not justify the accepted implication. Negative-cycle closure can discard a branch only after its exact emptiness proof. Nonempty branches are all checked with the actual rows, complete boxes and rational signs.

The minimum strengthened frame bound is

$$\frac{8442028677986126589157}{10485760000000000000000000} = 0.00805094592856 \dots > \ell,$$

at label 40. The minimum pair payment is M at $(406, 70)$. Consequently the current limiting component is the outer case. The slightly larger exact minimum of the same stored data is $16891485543266126589157/20971520000000000000000000$; we choose the simpler, strictly smaller uniform reward c .

4 Counting transport, reproducibility and scope

The sum of gap pressures is still $B = 404350/10^8$. The original AM energy certificate pays gain $67216841/10^8$. Sliding the nine-point frame preserves each pair budget at most two, total pair mass below sixteen, and endpoint cost $8c$. The near-pair reward exceeds $1/40 > c$; the majorant has mass below two. The smoothing loss remains $32d\varepsilon$. The Hermitian Gram argument retains off-line reflected pairs and every multiplicity [2, 4]. First take the height limit at fixed smoothing, then decrease the smoothing parameter. Its proved lossless transport yields the claimed proportion

$$\frac{67216841/10^8 - B}{1 - c} = \frac{66812491}{99194552}.$$

The same finite inputs yield the dyadic statement; the retained counting transport yields the cumulative statement. This proves Theorem 1.1, including its interpretation in actual zeta counts rather than a model spectrum.

The public files in the research repository are

```
scripts/am_joint_frame_pair_certificate.py
output/am-joint-frame-pair-certificate.json.
```

The default standard-library checker recomputes all 241 frames and all 2,399 pair domains from the public frozen data. An optional source check matches the 425 original entries and REG packs. Optimized Python is rejected because its removal of assertions would invalidate verification.

The preceding 941021/1397107 theorem has a complete ordinary Lean proof and independent kernel replay [5]. The present paper records the additional mathematical certificate. Formal verification of its new real-domain and counting proofs, when completed, belongs to its own additive namespace and receipts in the independent formalization repository; compilation, independent replay and website resource compliance are separate records. All preceding proofs, papers and submissions are retained.

References

- [1] The fixed original continuous PC8 certificate, commit d272437e, original Solution.lean.
- [2] Li Xiang, *Complete Payment of the Ninth-Span Square for Simple Critical Zeros*, 2026. Retained research snapshot in RH-Weil.
- [3] Li Xiang, *Strengthening the Low Frames of a Nine-Point Certificate for Simple Critical Zeros*, 2026.
- [4] Li Xiang, *Lossless Eight-Point Assembly for Simple Critical Zeros of the Riemann Zeta Function*, 2026.
- [5] Li Xiang, RH-Zero-Proportion-Formalization, c705c4a, complete NinthSpan Lean proof and independent nanoda replay, 2026.